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RESEARCH ARTICLE

Genetic diversity and divergence in the fountain darter (*Etheostoma fonticola*): implications for conservation of an endangered species

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Abstract The endangered fountain darter *Etheostoma fonticola* is found only in the Comal and San Marcos rivers in the Guadalupe River basin in central Texas, USA. Comal River fountain darters were believed to be extirpated following a severe drought in the 1950s and were reintroduced in the early 1970s using 457 darters from the San Marcos River. In this study we used 23 microsatellite loci to describe and evaluate the genetic diversity, population structure and effective population size (N_e) of fountain darters. We also evaluated the genetic effect of the Comal River reintroduction and the influence of low-head dams (dams) on dispersal in both rivers. Bayesian analysis of individual genotypes and Analysis of Molecular Variation supported two distinct populations concordant with the two rivers. Estimates of N_e were much smaller (<10 %) than census size (N_c) in both rivers but did not indicate the populations are at risk of an immediate and rapid loss of genetic diversity. Coalescent-based estimates of the genetically effective number of founders (*N*f) for the Comal River averaged about 49 darters and, together with the indices of genetic diversity and the bottleneck test (heterozygosity excess) results, were consistent with a founder event following the reintroduction in the Comal River. Finally, our results regarding the influence of dams on fountain darter dispersal were equivocal and did not support a conclusion. We recommend this issue be examined further as part of the fountain darter recovery program.

Keywords Fountain darter · Genetic diversity · Endangered species · Reintroduction · Effective founder number

Introduction

The fountain darter *Etheostoma fonticola* is a small, spring-endemic percid restricted to the upper San Marcos and Comal rivers in the Guadalupe River basin in central Texas, USA. The species is listed as endangered under the U.S. Endangered Species Act (ESA 1973, as amended) and captive refuge populations from each river have been established at the San Marcos Aquatic Resources Center (USFWS 1996). The fountain darter recovery plan calls for conducting biological studies necessary for successful monitoring, management and restoration (USFWS 1996). Among the studies needed to inform the recovery effort is a description and evaluation of genetic diversity in both the San Marcos and Comal rivers. In this context there is a need to develop a genetic baseline for on-going monitoring of the wild populations and for management of the captive refuge populations (USFWS 1996). There is also a need to evaluate restoration actions and other factors that may influence genetic diversity in fountain darters such as a past

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reintroduction event and low-head dams that may impede gene flow in both rivers.

One of the main threats to fountain darters is drought (natural and anthropogenic) and essential habitat for the fountain darter is strongly influenced by the amount of spring water emerging from the Edwards Aquifer at the headwaters of each river. The discharge is determined by the amount of precipitation over aquifer recharge areas and the amount of water extracted from the aquifer for human use. Droughts in central Texas generally occur at least once a decade and during a severe drought in the 1950s, Comal Springs ceased flowing for six months (Schenck and Whiteside 1976; Brune 2002). The fountain darter was likely extirpated from the Comal River during this drought (USFWS 1996) and physical separation prevented natural recolonization from the San Marcos population. In 1974–1976 457 darters from the San Marcos River were transplanted to the headsprings area of the Comal River (Schenck and Whiteside 1976; USFWS 1996). The reintroduction was successful in the sense that fountain darters are now found throughout the Comal River and the population size is estimated at over 150,000 fish (Linam et al. 1993; Ed Oborny 2011 pers. comm.). However, to inform restoration planning a genetic assessment of the reintroduction is needed to determine if the number of fountain darters used was sufficient to retain genetic diversity relative to the source (San Marcos) population and to evaluate the extent to which the two populations have diverged from one another.

Habitat fragmentation may also threaten fountain darters. Dams and other flow-control structures can have dramatic effects on lotic habitat by altering water chemistry and flow (Baxter 1977), river geomorphology (Ligon et al. 1995), fish and macroinvertebrate communities (Lessard and Hayes 2003; Santucci et al. 2005; Tiemann et al. 2004) and can cause a reduction of gene flow among populations, leading to reduced genetic diversity and increased genetic differentiation (Wofford et al. 2005; McCraney et al. 2010). Low-head dams (small drop structures in rivers, streams or channels constructed to impound water, meter discharge, or maintain stream slope; Kern et al. 2015) and weirs have been shown to influence gene flow and genetic diversity in some darter species (e.g., Haponski et al. 2007; Beneteau et al. 2009; Sterling et al. 2012). Low head dams (hereafter dams) in the Comal and San Marcos rivers impound water for recreational use and are a potential source of habitat fragmentation for fountain darters. Five dams in the Comal River and three dams in the San Marcos River may partially restrict gene flow within each river, resulting in genetic differentiation.

Here we use 23 microsatellite loci to describe and evaluate the genetic diversity, population structure and effective population size of fountain darters to inform restoration planning and provide a baseline for genetic monitoring using indices such as heterozygosity, allele number, *F*st and effective population size (Schwartz et al. 2007). We evaluate the genetic outcome of the reintroduction and the effect of dams on genetic diversity within and between the Comal and San Marcos rivers by addressing three questions: (1) have fountain darters in the Comal River diverged from fountain darters in the San Marcos River? (2) was the 457 fish reintroduction sufficient to maintain genetic diversity relative to the source population? (3) do dams impede gene flow? The results are assessed with regard to on-going and future conservation efforts including the selection and size of a refuge population(s) for ex situ captive propagation.

Methods

Sample collection

Fountain darters were collected from seven locations on the Comal River (n = 147) and nine locations on the San Marcos River (n = 180, Table 1; Fig. 1). The sample locations represent four sections in each river that are separated by dams that could inhibit gene flow (Fig. 1). Fish were anesthetized with tricane methane-sulfonate (MS-222; Finquel, Argent Chemical Laboratories, Inc., Redmond, Washington) and preserved individually in vials of 95 % ethanol. Fin tissue samples were collected for genetic analysis, preserved in 95 % ethanol, and sent to the Conservation Genetics Laboratory, Alaska Region, U.S. Fish and Wildlife Service for analyses.

Laboratory analyses

Twenty-three microsatellite loci were used to estimate genetic variation in the 327 fountain darter samples (Online Resource 1, Online Resource 2). Total genomic DNA was extracted from fin tissue (~25 mg) using proteinase K with the DNeasy™ DNA isolation kit (Qiagen Inc. Valencia, CA), quantified with fluorometry and diluted to a standard concentration. An MJResearch DNA Engine® thermal cycler was used to perform polymerase chain reactions (PCR) in 10 µl volumes; general conditions were: 2.5 mM $MgCl_2$, 1X PCR buffer (20 mM Tris–HCl pH 8.0, 50 mM KCl), 200 µM of each dNTP, 0.40 µM fluorescently labeled forward primer, 0.40 µM unlabeled reverse primer, 0.008 units *Taq* polymerase, and 1 µl of DNA (30 ng/µl). Standard thermal cycling conditions were: initial denaturation cycle of 92 °C for 2 min, followed by 92 °C for 15 s, 52–60 °C for 15 s (locus-specific sequences and annealing temperature are given in Online Resource 1), 72 °C for 30 s, (30 cycles) with a final single cycle of

Table 1 A summary of the samples, including location, date, collection designation, and sample size (n) used for genetic analysis of fountain darters (*E. fonticola*) from the Comal (CR) and San Marcos (SMR) rivers, Guadalupe River Basin, Texas, USA

River	Location	Label	n	Date	Collection
Comal	Houston St., Landa Lake	CR1a	15	02/25/10	CR1
Comal	Liberty Ave., Landa Lake	CR1b	15	02/25/10	CR1
Comal	Landa Lake, Landa Park	CR1c	9	03/09/10	CR1
			21	04/14/10	
Comal	Elizabeth Ave., Old Channel	CR2	1	02/25/10	CR2
			13	02/26/10	
			5	04/16/10	
			6	04/21/10	
Comal	Hinman Island Park, New Channel	CR3a	9	03/31/10	CR3
			12	04/09/10	
			9	04/14/10	
Comal	Above Hinman Weir, Old Channel	CR3b	2	04/14/10	CR3
Comal	Garden St.	CR4	10	03/31/10	CR4
			9	04/14/10	
			11	04/16/10	
San Marcos	Spring Lake, near hotel	SMR1a	15	11/23/09	SMR1
			15	02/05/10	
San Marcos	Spring Lake, near outflow dam	SMR1b	15	11/23/09	SMR1
			15	02/05/10	
San Marcos	Sewell Park	SMR2a	15	03/19/10	SMR2
			14	11/23/09	
San Marcos	City Park	SMR2b	15	03/17/10	SMR2
			15	04/09/10	
San Marcos	Rio Vista Park	SMR2c	14	03/17/10	SMR2
San Marcos	Cheatum St.	SMR3a	15	03/17/10	SMR3
San Marcos	I-35	SMR3b	15	03/19/10	SMR3
San Marcos	Cypress Tree, Lower	SMR4a	4	10/28/09	SMR4
San Marcos	Todd Island	SMR4b	3	10/28/09	SMR4
			4	04/09/10	
			6	04/23/10	

The location labels denote the river (CR, SMR), the river section (1, 2, 3, 4), and the location within the river section (a, b, c)

72 °C for 10 min. One-half µl of PCR product was electrophoresed and visualized with the Applied Biosystems 3730 Genetic Analyzer utilizing a polymer denaturing capillary system. Microsatellite allele sizes were estimated and scored by the computer program GeneMapper® version 4.0. Applied Biosystems GeneScan™-600 LIZ® size standards, 20–600 bases, were loaded in all lanes as an internal lane standard. Two independent researchers verified all scores manually, with discrepancies being resolved by replicating the analysis for the samples in question and repeating the double scoring process until scores matched. Data for samples from at least one column (n = 8) in each 96-well sample plate was automatically replicated to confirm that proper plate orientation was maintained throughout genotyping efforts for the project.

Genetic diversity and divergence

We examined population structure to assess if fountain darters in the Comal River have diverged from fountain darters in the San Marcos River and if dams in both rivers impede gene flow. We used complementary individual- and population-level analyses. Individual-level analysis was done using the software STRUCTURE v. 2.3.1 (Pritchard et al. 2000). STRUCTURE is a Bayesian, model-based algorithm that clusters genetically similar individuals and estimates the most likely clustering scenario, assuming Hardy–Weinberg and linkage equilibrium. We tested K = 2 to 16 clusters assuming admixture and correlated allele frequencies (between the K clusters) and using a burn-in of 20,000 replications followed by 50,000 Markov

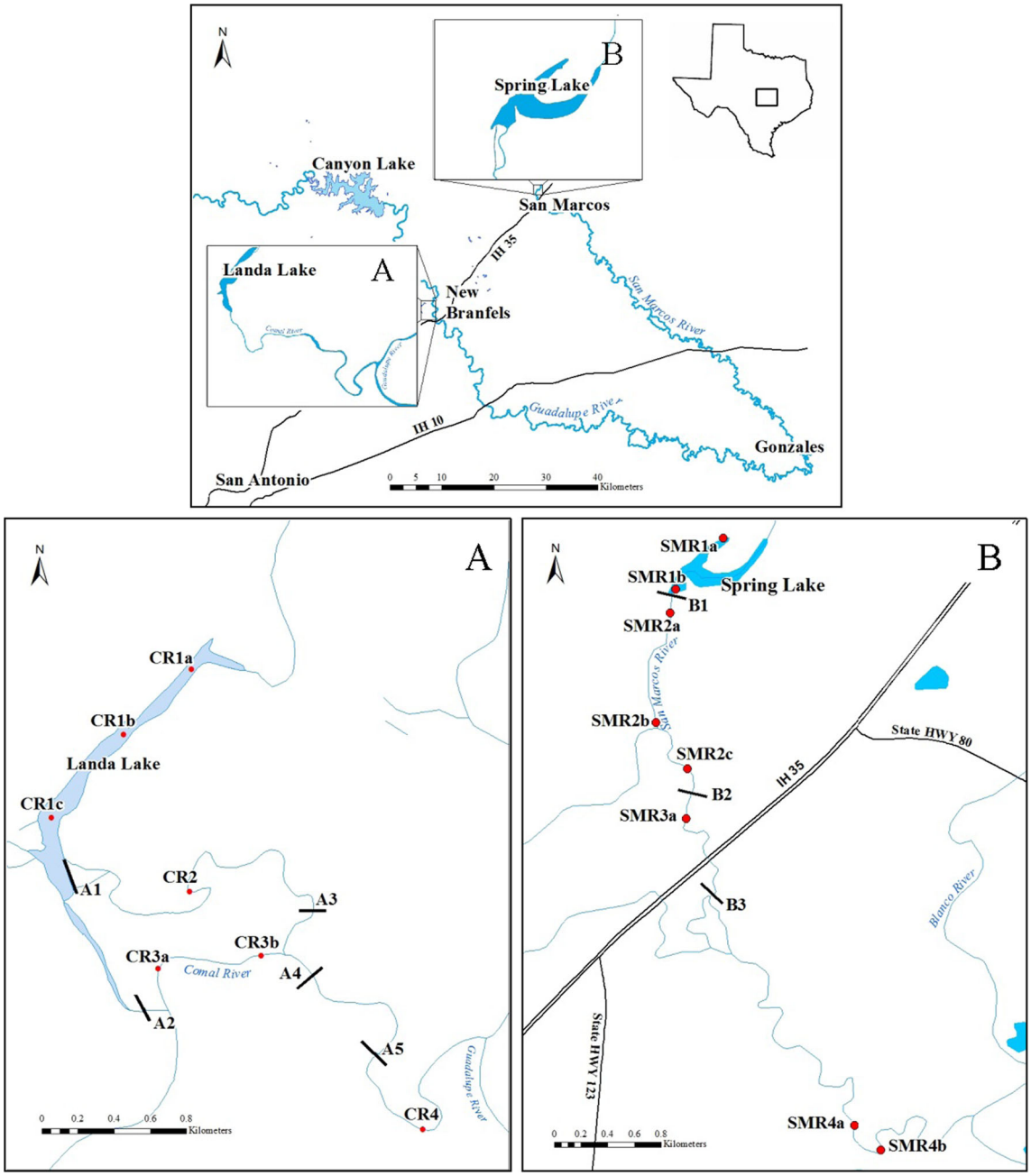

Fig. 1 A map showing the sample locations for fountain darters (*E. fonticola*) from the **a** Comal (CR) and **b** San Marcos (SMR) rivers, Guadalupe River Basin, Texas, USA. The 16 sites (*red circles*) are coded as in Table 1 where the *upper case letters* (CR, SMR) denote the river, the numbers (1, 2, 3, 4) denote the river section, and *lower case letters* (a, b, c) denote the location within the river section. The dams are coded (A = Comal River, B = San Marcos River) and represented by *straight black lines* perpendicular to river flow

chain Monte Carlo replications. Ten replications were performed to confirm the consistency of the log-likelihood probabilities and to estimate the variance. The most likely cluster scenario was the one that produced the largest penalized likelihood value (the mean of the likelihood values from each replicate minus half the variance, Pritchard et al. 2000) and the results were visualized using the software Distruct v. 1.1 (Rosenberg 2004).

For the population-level analyses, we grouped the samples into collections by river section (see collections, Table 1) to both evaluate the dams as putative barriers to gene flow and evaluate genetic divergence between the two rivers. The analyses included pairwise tests of genetic differentiation and tests of hierarchical structure. The software GenePop v. 4.1.4 (Rousset 2008) was used to conduct exact *G*-tests of allele frequency homogeneity to

test for genetic differentiation between all pairs of collections within and between each river. Significant *G*-test results for adjacent collections would be consistent with restricted gene flow by the dams. The *p* value for each test was estimated using a Markov chain algorithm with the following parameter values: dememorisation = 10,000, batch length = 500, batch number = 5000. The software FSTAT v. 2.9.3 (Goudet 2001) was used to calculate pairwise estimates of *F*st (Wright 1943), according to Weir and Cockerham (1984), and quantify the relative level of genetic differentiation between all pairs of collections. Analysis of Molecular Variation (AMOVA) was used to compute hierarchical F-statistics and quantify the relative level of genetic variation among all collections (*F*st), among collections between rivers (*F*ct) and among collections within rivers (*F*sc) and to test for statistical significance of each value. We performed AMOVA using the software ARLEQUIN v. 3.5 (Excoffier et al. 2005).

Estimates of allele frequency, allelic richness (*A*r) and observed (*H*o) and expected heterozygosity (*H*e) were computed for each locus and collection using FSTAT. Randomization tests were used to test for conformity to Hardy–Weinberg expectations (HWE) for each locus and collection combination and to test for composite genotypic disequilibrium among locus pairs over all collections. These tests were performed using FSTAT and the threshold for statistical significance ($\alpha = 0.05$) was corrected (α/k) using the sequential Bonferroni method (Rice 1989) for k simultaneous tests where the value of k decreased sequentially by removing significant tests until no tests were judged significant. Two initial values of k were used for the HWE test to evaluate each collection over all loci ($k = 23$) and each locus over all collections ($k = 8$).

Effective population size and genetic evaluation of the Comal River reintroduction

The analysis of effective population size (*N*e) and genetic evaluation of the Comal River reintroduction were each performed after pooling collections for each river because we found little evidence of population structure within rivers. The contemporary N_e of fountain darters in the Comal and San Marcos rivers was estimated using the linkage disequilibrium method implemented in the software N_EESTIMATOR 2.01 (Do et al. 2014). The estimates were derived with $P_{crit} = 0.02$ (the minimum frequency for alleles to be included in the analysis) and confidence intervals were generated using the jackknife method.

To evaluate if the reintroduction was sufficient to maintain genetic diversity relative to the source population we tested if genetic diversity in fountain darters from the Comal River was lower than in fountain darters from the San Marcos River. We computed estimates of *H*e, *A*r and private allele richness (pAr) for each locus in each population using the software HP-RARE v. 1.0 (Kalinowski 2005). A Wilcoxon paired-sample test implemented in the software R v. 3.2.2 (http://www.r-project.org/) was used to test the null hypothesis that, for each statistic, estimates across loci from the Comal population were greater than or equal to estimates from the San Marcos population.

The genetically effective number of individuals re-introduced to the Comal River were estimated from the allele frequency data using a coalescent-based maximum likelihood approach in the software COALIT and NFCONE (Anderson and Slatkin 2007). This method requires estimates of three parameters: the intrinsic rate of population growth (R), the number of generations since introduction (T), and the effective carrying capacity (Nk). We used multiple estimates for each parameter to reflect a reasonable range of values based on knowledge of fountain darter biology and habitat. For example, we examined estimates of 35 and 70 for T because fountain darters may reproduce one to two times per year (USFWS 1996) and the reintroduction occurred 35 years before the samples were collected. We examined estimates of 0.5, 1.0, 2.0 and 3.0 for R because fountain darters rapidly expanded their range following the reintroduction at Landa Lake and are now found throughout the Comal River (Fig. 1a). We examined estimates of 5000, 50,000 and 100,000 for Nk because the effective carrying capacity likely varies as a result of drought. Estimates of the effective number of founders (*N*f) were derived for all 24 combinations of R, T, and Nk. The number of simulated coalescent trees computed by COALIT was 100,000 and the number of sampling replicates used by NFCONE was 10,000.

Finally, we evaluated the data for evidence of a genetic bottleneck in the Comal River. The software Bottleneck v. 1.2.02 (Cornuet and Luikart 1996) was used to test if the heterozygosity computed from observed allele frequencies was greater (heterozygote excess) than the equilibrium heterozygosity computed from allele frequencies expected at drift-mutation equilibrium. This excess of heterozygosity may be observed in a population following a bottleneck because allelic diversity is reduced faster than heterozygosity during a bottleneck. The analysis was run assuming a two-phase mutation model with 30 % multi-step mutations and a variance of 30 for the number of repeat units per multi-step mutation. The simulation included 5000 iterations and the results were evaluated using the sign test and standardized differences test.

Results

Genetic diversity and divergence

The analysis of individual genotypes using STRUCTURE indicated that the most likely solution was K = 2 clusters.

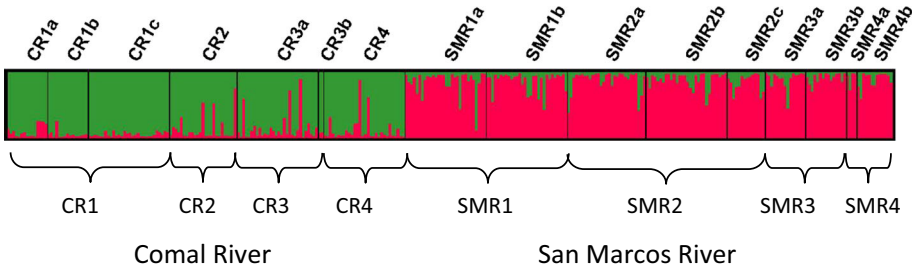

Fig. 2 A bar chart from the STRUCTURE analysis showing the most likely cluster scenario (K = 2) for fountain darters (*E. fonticola*) from the Comal and San Marcos rivers, Guadalupe River Basin, Texas, USA. Each individual is represented as a *vertical line* and the *color* indicates the proportion of the individual genotype from the Comal River cluster (*green*) and San Marcos River cluster (*red*). The *black vertical lines* separate the sample locations (top labels) and the *horizontal brackets* denote the collections as in Table 1

These two clusters aggregated individuals by river (Fig. 2). No evidence of genetic divergence was found using STRUCTURE when the Comal and San Marcos river samples were analyzed separately (the hierarchical method, e.g. Vähä et al. 2007) to assess if fine-scale population structure existed within each river. Nineteen of 28 paired collections showed statistically significant differences in allele frequencies when the *G*-test *p*-values were adjusted for multiple tests (Table 2). Sixteen of the significant results occurred when the paired collections were from different rivers. Three of the Comal River collections (CR1, CR2, CR4) showed significantly different allele frequencies from one another but not from the CR3 collection. Collections CR1 and CR2 were adjacent and separated by dam A1 (Fig. 1a). The San Marcos River collections were not significantly different from one another. The estimates of genetic variation among collections as measured by pairwise *F*st ranged from −0.0008 (SMR3 × SMR4) to 0.0209 (CR1 × SMR3) and were generally largest when the collections were from different rivers (Table 2). The AMOVA estimate of *F*st over all collections was 0.0142 (Table 3). The estimates of genetic variation between the two rivers (*F*ct) and among collections within rivers (*F*sc) were 0.0102 and 0.0040, respectively. The estimates of *F*st, *F*ct and *F*sc were all significantly greater than zero.

Table 3 A summary of the AMOVA results for fountain darters (*E. fonticola*) from the Comal and San Marcos rivers, Guadalupe River Basin, Texas, USA

Among all collections	*F*st	0.0142*
Among collections within rivers	*F*sc	0.0040*
Among rivers	*F*ct	0.0102*

The analysis was performed using the collections described in Table 1 and the estimates indicate the level of population structure among all collections (*F*st), among collections within the two rivers (*F*sc), and between the two rivers (*F*ct). An asterisk indicates the value is significantly larger than zero ($p < 0.05$)

The estimates of genetic diversity within collections as measured by average heterozygosity (*He*) and allelic richness (*Ar*) were lowest in the Comal River collection CR4 at 0.624 and 6.99 and highest in the San Marcos River collection SMR4 at 0.649 and 8.61 (Table 4). Tests of Hardy–Weinberg equilibrium initially revealed a deficit of heterozygote genotypes ($p < 0.05$) at 13 of the 184 paired locus x collection tests (Table 4). Six of these test were judged statistically significant after correction for multiple

Table 2 A summary of the pairwise estimates of *F*st (below diagonal) and the *p*-values from the *G*-test of allele frequency homogeneity (above diagonal) for fountain darter (*E. fonticola*) sample collections (Table 1) from the Comal (CR) and San Marcos (SMR) rivers, Guadalupe River Basin, Texas, USA

Aggr	Comal River				San Marcos River			
	CR1	CR2	CR3	CR4	SMR1	SMR2	SMR3	SMR4
CR1	NA	**0.0007**	0.1370	**≤0.0001**	**≤0.0001**	**≤0.0001**	**≤0.0001**	**≤0.0001**
CR2	0.0067	NA	0.3718	**0.0013**	**≤0.0001**	**≤0.0001**	**≤0.0001**	**≤0.0001**
CR3	0.0029	0.0028	NA	0.0558	**≤0.0001**	**≤0.0001**	**≤0.0001**	**≤0.0001**
CR4	0.0129	0.0082	0.0068	NA	**≤0.0001**	**≤0.0001**	**≤0.0001**	**≤0.0001**
SMR1	0.0114	0.0113	0.0099	0.0142	NA	0.2331	0.0062	0.4161
SMR2	0.0168	0.0145	0.0126	0.0133	0.0004	NA	0.1080	0.2361
SMR3	0.0209	0.0155	0.0119	0.0196	0.0031	0.0017	NA	0.2435
SMR4	0.0141	0.0139	0.0099	0.0134	0.0007	−0.0002	−0.0008	NA

A bold and underlined *p* value indicates a statistically significant difference in allele frequencies using the sequential Bonferroni method (28 simultaneous tests, initial alpha = 0.002)

tests, however these results were not isolated to a single locus or collection.

Effective population size and genetic evaluation of the Comal River reintroduction

The estimates of contemporary N_e with 95 % confidence intervals for the Comal and San Marcos populations were 899 (461—7668) and 9234 (1908—inf). The Wilcoxon paired-sample test showed that the estimates of *He*, *Ar* and pAr were all significantly larger across loci ($p < 0.05$) in the San Marcos population compared to the Comal population (Table 5). Seventy-five private alleles were observed in the San Marcos population and nine private alleles were observed in the Comal population (Online Resource 3). The mean frequency of private alleles was 0.010 (SD = 0.006) for the San Marcos population and 0.007 (SD = 0.005) for the Comal population.

The maximum likelihood estimates of the effective number of founders (*N*f) ranged from 37.21 to 77.42 for the 24 scenarios using combinations of intrinsic rate of population growth (R), the number of generations since introduction (T), and the effective carrying capacity (Nk, Fig. 3). One scenario (R = 0.5, T = 70, Nk = 5000) resulted in a negative estimate of Nf and was not included in the evaluation. The mean estimate over all scenarios (49.35) suggests that the effective number of founders for the Comal River was approximately 10.8 % of the number of fish (457) transferred from the San Marcos River. The Bottleneck test results showed 17 of 23 loci with an excess of heterozygosity (relative to the expected heterozygosity at drift-mutation equilibrium). An excess of heterozygosity may be observed in a population following a bottleneck because allelic diversity is lost much faster than heterozygosity. The results of the sign test ($p = 0.062$) and standardized differences test ($p = 0.085$) provided weak evidence for a bottleneck.

Discussion

Genetic diversity and divergence

Two results stand out from our evaluation of genetic diversity and divergence in the endangered fountain darter. First, the individual-level Bayesian clustering analysis of genotypes supported two populations represented by the Comal and San Marcos rivers. The two populations are separated by over 200 river kilometers along the mainstem Guadalupe River and lower San Marcos River (Fig. 1). This distance, the lack of suitable habitat between the populations, and the high site fidelity in the species, is thought to prevent fountain darter migration between the two rivers. For example, Dammeyer et al. (2013) found that fountain darters moved on average 10 ± 17 m throughout the year during a stable hydrologic regime, and movement tended to be upstream and driven by habitat variables such as low growing aquatic vegetation and stream flow. Many darter species have limited dispersal ability with specific habitat requirements that result in reproductively isolated populations exhibiting substantial differentiation on a small geographic scale (Haponski et al. 2009; Robinson et al. 2013; Fitzpatrick et al. 2014). The threatened Okaloosa darter (*Etheostoma okaloosae*), for example, is restricted to a 457 km^2 watershed in Florida and AMOVA results from microsatellite loci revealed a high level of differentiation ($\Phi_{PR} = 0.334$) among collections from different drainages (Austin et al. 2011). Similarly, the population-level AMOVA results from this study indicated that genetic divergence in fountain darters is largely due to differences among collections from the two rivers. It is worth noting that the level of differentiation between the Comal and San Marcos rivers (*F*ct = 0.0102) is, by comparison, low but significant and shows that the reintroduced Comal population has diverged at neutral loci from the San Marcos source population.

Our population-level evaluation also revealed very little evidence of genetic divergence among collections within the two rivers. River barriers including dams and weirs have been shown to influence dispersal of some darter species (Haponski et al. 2007; Beneteau et al. 2009). However, we only found evidence of genetic differentiation at one dam (A1 in the Comal River) between two adjacent fountain darter collections (CR1 × CR2). Indeed, downstream movement is clearly evident in the Comal River where fountain darters were reintroduced in the Landa Lake area near the headwaters but now occupy the entire length (5 km) of the river (Fig. 1). Collectively, our results regarding the influence of dams on fountain darter dispersal are equivocal and do not provide strong evidence that movement across these barriers is inhibited. This outcome is surprising but may indicate that periodic opportunities arise, such as occasional flooding events, for movement either across or around these dams. Another possible explanation is that number of fountain darters above and below the dams is large enough to minimize the rate of genetic drift and thus limit the power to detect genetic differentiation (Lowe and Allendorf 2010).

Effective population size and genetic evaluation of the Comal River reintroduction

Two results stand out regarding the contemporary N_e estimates. First, the estimates for both populations, while larger than those reported for some other darter species (e.g., Fluker et al. 2010; Fitzpatrick et al. 2014), are much

Table 4 A summary of genetic diversity estimates for 23 microsatellite loci in eight fountain darter (*E. fonticola*) collections from the Comal (CR) and San Marcos (SMR) rivers, Guadalupe River Basin, Texas, USA

Locus	Stat	Comal River				San Marcos River			
		CR1 (60)	CR2 (25)	CR3 (32)	CR4 (30)	SMR1 (60)	SMR2 (73)	SMR3 (30)	SMR4 (17)
Eche001	*He*	0.546	0.381	0.607	0.346	0.424	0.313	0.505	0.524
	Ho	0.533	0.400	0.531	0.333	0.433	0.288	0.500	0.471
	Ar	3.26	2.68	3.78	3.63	3.61	3.49	3.97	4.00
Eche002	*He*	0.307	0.256	0.177	0.129	0.187	0.143	0.128	0.404
	Ho	**0.233**	0.240	0.188	**0.067**	0.167	0.123	0.133	0.294
	Ar	3.18	2.96	3.02	2.49	2.73	2.64	2.49	4.00
EosC112	*He*	0.000	0.000	0.031	0.000	0.017	0.027	0.033	0.000
	Ho	0.000	0.000	0.031	0.000	0.017	0.027	0.033	0.000
	Ar	1.00	1.00	1.53	1.00	1.28	1.47	1.57	1.00
EosC2	*He*	0.342	0.528	0.478	0.582	0.405	0.523	0.452	0.557
	Ho	0.350	0.600	0.469	0.533	0.400	0.479	0.467	0.588
	Ar	2.64	3.00	2.99	3.00	2.96	2.95	2.00	3.00
EosC3	*He*	0.325	0.431	0.441	0.357	0.382	0.407	0.506	0.467
	Ho	0.317	0.440	0.438	0.367	0.400	0.425	0.600	0.588
	Ar	2.82	2.90	2.78	2.82	2.28	2.66	2.57	2.00
EosC6	*He*	0.835	0.762	0.829	0.783	0.887	0.881	0.851	0.875
	Ho	0.817	0.720	0.875	0.667	0.917	0.890	0.833	0.882
	Ar	7.05	7.26	6.96	7.02	10.22	9.90	9.33	9.00
Esc132b	*He*	0.937	0.928	0.957	0.940	0.956	0.957	0.959	0.949
	Ho	0.950	1.000	0.969	0.933	0.967	0.932	0.967	0.941
	Ar	16.02	16.70	18.79	16.59	18.35	19.15	19.81	18.00
Esc26b	*He*	0.872	0.885	0.903	0.881	0.913	0.925	0.925	0.932
	Ho	0.883	0.840	0.938	0.933	**0.817**	0.890	0.833	0.882
	Ar	10.03	11.81	11.67	9.33	13.13	13.27	13.82	14.00
Esc57	*He*	0.017	0.000	0.000	0.000	0.000	0.000	0.000	0.059
	Ho	0.017	0.000	0.000	0.000	0.000	0.000	0.000	0.059
	Ar	1.28	1.00	1.00	1.00	1.00	1.00	1.00	2.00
EfoA12	*He*	0.466	0.548	0.507	0.445	0.415	0.372	0.474	0.412
	Ho	0.450	0.440	0.438	0.400	0.367	0.411	0.533	0.412
	Ar	4.37	5.47	4.68	3.94	4.93	4.80	5.43	4.94
EfoD4	*He*	0.897	0.892	0.910	0.918	0.915	0.912	0.914	0.930
	Ho	0.948	1.000	0.969	0.967	0.898	0.904	0.900	0.941
	Ar	11.43	11.68	12.13	12.50	13.19	12.19	13.07	13.70
EfoE3	*He*	0.917	0.930	0.923	0.934	0.930	0.931	0.928	0.947
	Ho	0.850	**0.800**	0.903	0.867	0.900	0.918	0.933	0.824
	Ar	13.04	13.66	13.46	13.32	14.68	14.08	14.04	15.58
EfoE12	*He*	0.711	0.683	0.737	0.751	0.748	0.685	0.645	0.735
	Ho	0.700	0.640	0.688	0.767	0.733	0.671	0.567	0.706
	Ar	5.64	5.59	5.74	4.98	5.22	5.30	5.95	5.88
EfoE109	*He*	0.096	0.273	0.119	0.259	0.224	0.249	0.263	0.231
	Ho	0.100	0.320	0.125	0.300	0.250	0.282	0.300	0.250
	Ar	1.85	2.00	1.94	2.00	2.45	2.53	2.53	3.00
EfoE116	*He*	0.269	0.255	0.228	0.216	0.228	0.287	0.343	0.268
	Ho	0.283	0.200	0.250	0.200	0.250	0.329	0.367	0.177
	Ar	3.12	2.87	2.74	2.77	2.94	3.02	2.96	2.94
EfoE125	*He*	0.806	0.838	0.839	0.793	0.855	0.852	0.890	0.869

Table 4 continued

Locus	Stat	Comal River				San Marcos River			
		CR1 (60)	CR2 (25)	CR3 (32)	CR4 (30)	SMR1 (60)	SMR2 (73)	SMR3 (30)	SMR4 (17)
	Ho	0.850	0.760	1.000	0.867	0.900	0.822	0.833	0.765
	Ar	7.17	6.63	8.25	6.83	9.00	8.91	10.68	9.82
EfoE134	*He*	0.902	0.892	0.866	0.910	0.946	0.938	0.953	0.952
	Ho	**0.717**	0.920	0.781	0.900	**0.850**	0.889	0.867	**0.813**
	Ar	12.25	11.87	11.34	12.85	16.13	15.50	16.17	14.00
EfoE150	*He*	0.893	0.920	0.903	0.876	0.916	0.922	0.929	0.938
	Ho	0.917	0.880	0.936	0.900	**0.783**	0.904	0.900	0.941
	Ar	11.81	12.99	12.82	9.74	15.04	15.07	14.76	16.46
EfoE159	*He*	0.793	0.779	0.807	0.812	0.787	0.800	0.805	0.827
	Ho	**0.617**	0.640	0.813	0.733	**0.583**	**0.667**	0.767	0.813
	Ar	4.99	5.75	5.50	5.53	5.46	6.27	5.94	7.00
EfoE185	*He*	0.924	0.894	0.912	0.918	0.940	0.939	0.916	0.947
	Ho	0.933	1.000	0.903	0.833	0.950	0.959	0.967	0.882
	Ar	13.92	12.56	12.95	11.96	16.16	15.17	14.89	16.47
EfoE189	*He*	0.770	0.768	0.786	0.726	0.821	0.809	0.809	0.792
	Ho	0.800	0.760	0.656	0.733	0.783	0.819	0.767	0.765
	Ar	7.09	7.07	6.88	7.20	9.08	8.59	7.95	8.82
EfoE192	*He*	0.902	0.864	0.874	0.911	0.939	0.931	0.928	0.921
	Ho	0.883	0.840	**0.719**	**0.800**	0.883	0.904	0.833	1.000
	Ar	11.77	10.57	11.49	12.07	14.65	14.25	13.70	13.65
EfoE193	*He*	0.901	0.900	0.878	0.856	0.894	0.890	0.887	0.884
	Ho	0.917	0.880	0.906	0.800	0.950	0.863	0.833	0.941
	Ar	10.10	11.20	10.01	9.01	10.94	10.95	11.14	9.88
Avg	*He*	0.627	0.635	0.640	0.624	0.640	0.639	0.654	0.670
	Ho	0.612	0.623	0.631	0.604	0.617	0.626	0.641	0.649
	Ar	7.17	7.31	7.45	6.99	8.45	8.35	8.46	8.61

The estimates include expected heterozygosity (*He*), observed heterozygosity (*Ho*), and allelic richness (*Ar*). Estimates of *Ho* lower than expected based on Hardy–Weinberg expectations are underlined and in bold ($p < 0.05$). The sample size of each collection is in parentheses

Table 5 A summary of estimates of mean heterozygosity (*He*) mean allelic richness (*Ar*) and mean private allele richness (pAr) across loci for fountain darter (*E. fonticola*) from the Comal and San Marcos rivers, Guadalupe River Basin, Texas, USA

Population	*He*	*Ar*	pAr
San Marcos	0.645	13.47	3.05
Comal	0.634	10.90	0.48
p value	0.022	<0.001	<0.001

The *p* values are from the Wilcoxon paired-sample test of the null hypothesis Comal ≥ San Marcos for each statistic across all loci

smaller than the estimates of census size (N_c). Current census size estimates suggest each river supports at least 150,000 fish and thus the N_e/N_c ratios are 0.06 for the San Marcos River ($N_e = 9{,}234$) and 0.006 for the Comal River ($N_e = 899$) populations. While these ratios are below the average (0.11) reported by Frankham (1995) from a survey of 102 species, they are not unusual (e.g., Hauser et al. 2002). Frankham (1995) found that fluctuations in census population size is the most important factor influencing N_e across species and, given that the fountain darter habitat is exposed to periodic drought, it is likely that this species exhibits large shifts in N_c as habitat is lost and restored. The second notable result is that the effective size of the San Marcos population is about 10.3 times larger than the Comal population. The restored Comal population was founded from a small effective number of fish from the San Marcos population (see below). Such recent bottlenecks can downwardly bias estimates of contemporary N_e using the linkage disequilibrium method (Waples 2005). Other factors could also influence the difference in N_e between the two populations including differences in carrying capacity of the two rivers and differences in the influence of drought on available habitat. Further research should examine these and other factors that may influence N_e in fountain darters.

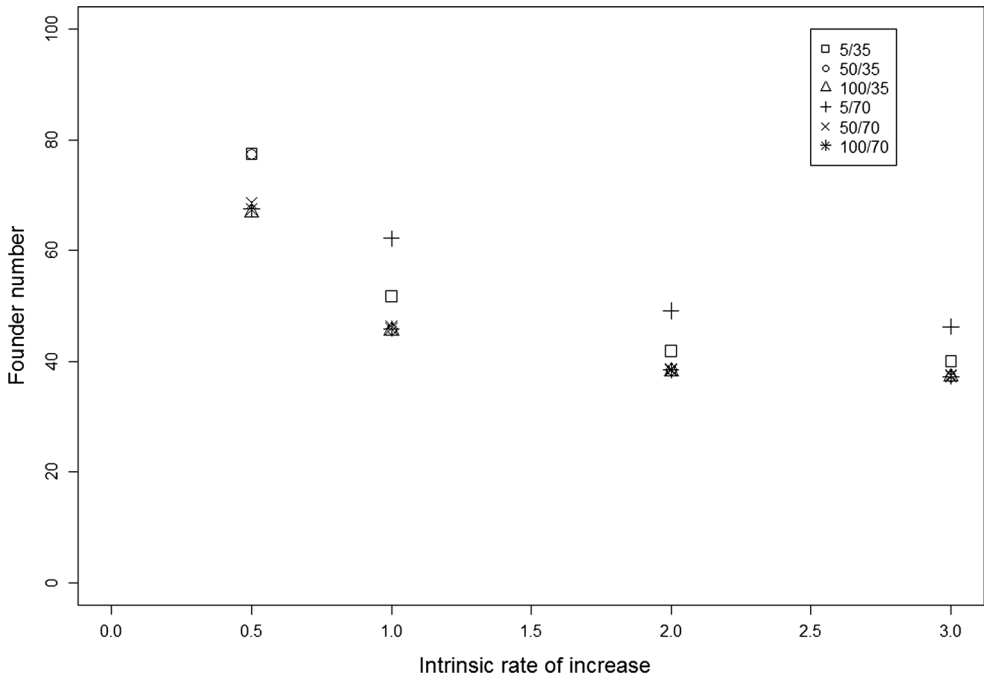

Fig. 3 A scatter plot showing the maximum likelihood estimates of the effective number of founders for fountain darters (*E. fonticola*) in the Comal River, Guadalupe River Basin, Texas, USA. The estimates were derived for 24 scenarios using three values for the effective carrying capacity (NK = 5,000, 50,000, 100,000), four values for the intrinsic rate of population growth (R = 0.5, 1.0, 2.0, 3.0), and two values for generations since founding (T = 35, 70). Each *symbol* represents one of the three values of NK (×1000) and one of the two values of T

Populations established as the result of a reintroduction often exhibit reduced genetic variability (Leberg 1990). Our results comparing estimates of heterozygosity, allelic richness and private alleles indicate that the reintroduced Comal population has lost genetic diversity relative to the San Marcos source population. Assuming the genetic diversity in the 457 fountain darters reintroduced into the Comal River was representative of the San Marcos population, our evaluation suggests that much of the observed loss of diversity and population divergence occurred soon after the reintroduction (founder effect) although the populations continue to diverge as a result of genetic drift. Rapid divergence at neutral loci is consistent with a founder event and has been documented in other small freshwater fishes following colonization (e.g., Kinziger et al. 2011). In the present study, the mean estimate of the effective number of founders ($Nf = 49.35$) was small compared to the number of fish ($N_c = 457$) used for the reintroduction. The Nf/N_c ratio (0.108) was close to the average (0.11) across taxon reported by Frankham (1995). We used the equation:

$$H_t/H_0 = [1 - 1/(2N_e)]^t,$$

(Frankham et al. 2002) to estimate the duration of the bottleneck in generations (t) assuming estimates of heterozygosity at time t (H_t) from the Comal population (0.634), initial heterozygosity (H_0) from the San Marcos population (0.645) and N_e from the effective founder number (49.35). The result suggests that the observed loss of heterozygosity in the Comal population relative to the San Marcos population could have occurred in as few as t = 1.8 generations. In addition, the signature of a founder event (heterozygosity excess) was evident in 17 of the 23 loci although the bottleneck tests provided weak statistical support ($0.05 < P < 0.10$). These tests are sensitive to the duration of the bottleneck and it is likely that, given the current N_c, the Comal River reintroduction was followed by a period of rapid population growth that has dampened the bottleneck signal (Hundertmark and Van Daele 2010; Peery et al. 2012).

The Comal population exhibited nine alleles not found in the San Marcos population (private alleles). It is possible that these private alleles originated from fountain darters that survived the drought in the early 1950s and that remained undetected despite the extensive survey effort prior to the reintroduction in 1974–76 (Schenck and Whiteside 1976; USFWS 1996). However, we believe the results suggest that this possibility is unlikely. For example, the mean private allele richness per locus and mean frequency of private alleles were both very low in the Comal population (0.48, 0.007) compared to the San Marcos population (3.05, 0.010). Remnant Comal River fountain darters would likely have possessed a higher number of private alleles than observed in the present Comal River sample. In addition, the relatively low level of genetic differentiation ($Fst = 0.0142$) between the two populations is consistent with recent divergence from genetic drift following the reintroduction and rapid bottleneck as opposed to a long-standing (pre-reintroduction) separation that included a long-term (>20 generations)

bottleneck in the Comal population following the drought in the 1950s. We believe, given the small number of private alleles in the Comal population sample, it is more likely that these alleles were simply not sampled from the San Marcos population or that they arose naturally through mutation given the relatively high mutation rate of microsatellites (Jarne and Lagoda 1996) and short generation time ($\sim$1 year) of fountain darters.

Conservation implications

The results from this study have implications with respect to conservation of the federally endangered fountain darter. First, the genetic data revealed two populations, the Comal and San Marcos populations, with no evidence of population structure within the San Marcos River and only weak evidence of population structure within the Comal River. Presently, fountain darters in both rivers are treated separately for conservation planning including ex situ captive propagation of two refuge populations at the San Marcos Aquatic Resources Center (USFWS 1996). Our results support this approach and captive propagation of the San Marcos population but indicate the need to re-evaluate captive propagation of the Comal population. That is, our results are consistent with the Comal population being descendent from the San Marcos population and thus it may only be necessary to maintain the source population (San Marcos) in captivity. It is important to note, however, that our genetic results are from neutral loci and we make no inference regarding local adaptation by reintroduced fountain darters in the Comal River. Evidence for rapid adaptation has been shown in other freshwater fish species. Stockwell and Vinyard (2000) found divergence of genetic-based life history characteristics in the western mosquitofish *Gambusia affinis*, after approximately 60 years. Further study is needed to assess if the Comal and San Marcos river fountain darters differ at adaptively important traits and genes.

The rate at which genetic diversity is lost is related to N_e not N_c and thus N_e is an important parameter to monitor in populations of conservation concern (Schwartz et al. 2007). The estimates of N_e here do not suggest these populations are at risk of an immediate and rapid loss of genetic diversity (Frankham et al. 2002). However, the fact that the estimates of N_e are much smaller than N_c for both populations warrants attention and highlights the importance of including N_e, as well as *He* and *Ar*, as part of a genetic monitoring program for fountain darters in order to fully evaluate the genetic consequences of reductions in census size following events such as droughts. Similarly, the estimates of the effective number of founders used in the Comal River reintroduction (range 37.21–77.42, mean = 49.35) were well below the number of fish (457) used for the reintroduction. These results suggest that future reintroductions into either river will require a larger sample ($\gg$457) from the captive refuge populations in order to avoid a founder event that results in a rapid loss of genetic diversity relative to the source population. Finally, while this study did not support a conclusion regarding the influence of dams on fountain darter gene flow, the issue is a critical one for conservation planning relative to water use in both rivers (USFWS 1996). Dammeyer et al. (2013) showed that fountain darters tend to move upstream and this behavior may be disrupted by the dams. For this reason we suggest that the influence of dams on fountain darter movement continue to be explored using a combination of genetic and ecological studies targeting fish adjacent to dams.

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